

7 Equilibria

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A dynamic equilibrium has equal forward and reverse rates in a closed system, though concentrations need not be equal. K_c uses equilibrium concentrations raised to stoichiometric powers; omit pure solids/liquids. Changing temperature changes K_c ; changing concentration or pressure shifts composition while K_c stays fixed at fixed temperature.

Worked example: follow the reasoning

For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $K_c = [\text{NH}_3]^2/([\text{N}_2][\text{H}_2]^3)$; increasing pressure favours fewer gas molecules.

Quick check · multiple choice

A What changes K_c for a fixed reaction?

[1]

- A Adding more solid
- B Pressure at fixed T
- C Temperature
- D A catalyst at fixed T

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $\text{A} + \text{B} \rightleftharpoons 2\text{C}$, equilibrium concentrations are $[\text{A}]=0.20$, $[\text{B}]=0.50$, $[\text{C}]=0.40 \text{ mol dm}^{-3}$. Calculate K_c .

[2]

Final answer: _____

Answers and explanation

Quick check: C. Temperature

Question 1 [2 marks]

$$K_c = [C]^2 / ([A][B]) \text{ [1]} = 0.40^2 / (0.20 \times 0.50) = 1.6 \text{ [1]}.$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.